

Bilal Amin

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EDUCATION

Universität des Saarlandes

Master's degree, Data Science and AI
Grade: 1.9 (German grading system)

Apr. 2024 – Present

Saarbrücken, Germany

University of Bergen (Erasmus+ Exchange Semester)

Exchange Student, Computer Science

Jan. 2026 – Jun. 2026

Bergen, Norway

National University of Sciences & Technology

Bachelor's in Mechatronics Engineering (Computer Science)
Grade: 1.7 (German grading system)

Sep. 2019 – Sep. 2023

Islamabad, Pakistan

EXPERIENCE

Research Assistant (HiWi) – Software Engineering

Chair of Software Engineering, Saarland University

Starting Oct. 2026

Saarbrücken, Germany

- Will work directly with Qingqing Dong on AI-assisted software engineering and model-driven system engineering research.
- Planned work includes investigating AI-assisted model merging, semantic merge-conflict detection, and feature-based merge recommendations.
- Will support research on software evolution and intelligent methods for improving complex software and model-based systems.

Student Assistant (HiWi) – Web Development

EuroComCenter, Saarland University

Sep. 2026 – Present

Saarbrücken, Germany

- Maintain and improve the EuroComCenter web platform, focusing on usability, multilingual content, and website structure.
- Review the existing implementation, fix technical and user-interface issues, and manage CMS-based content and digital media.
- Work with the project team to translate requirements into practical technical and design improvements.

Working Student – Internet Architecture Group

Max Planck Institute for Informatics (MPI)

Oct. 2024 – Mar. 2025

Saarbrücken, Germany

- Designed an automated framework to evaluate congestion control algorithms in NS-3 (Network Simulator).
- Created a user-friendly dashboard to visualize performance metrics under different network conditions.
- Developed system tests and metrics for performance evaluation across varied scenarios.
- Analyzed simulation results to identify optimization opportunities and guide design decisions.

Data Analyst

Convergent Business Technologies

Oct. 2023 – Mar. 2024

Islamabad, Pakistan

- Developed reproducible data science workflows using Python and SQL to manipulate and analyze data.
- As part of training, completed the CS50 course and created several dashboards related to the coursework.

Web Development Intern

CureMD

Jun. 2023 – Aug. 2023

Lahore, Pakistan

- Designed user interfaces using HTML, CSS, and JavaScript, improving usability and accessibility for end users.
- Developed APIs in the .NET framework, enhancing data integration and tool development.

Technical Intern

STEP (Space Technology Education and Popularization)

Aug. 2021 – Oct. 2021

Islamabad, Pakistan

- Led workshops teaching programming fundamentals using Scratch, focusing on hands-on learning for underprivileged youth.
- Created educational materials to simplify complex technology concepts, fostering interest in STEM fields.

PROJECTS

GermanLyft – AI-Powered German Learning Platform

2026

- Developed a full-stack German-learning platform using React, TypeScript, Flask, and the Groq LLM API, mainly for CEFR A1–A2 learners.
- Built an interactive AI tutor for grammar explanations, sentence correction, vocabulary practice, conversational learning, and dynamically generated exercises.
- Implemented placement tests, quizzes, user accounts, profiles, and Anki .apkg deck importing using a REST-based React/Flask architecture.
- Deployed the frontend on Vercel and the backend on Render.

AI Learning Assistant using LLMs (RAG)

2025

- Built an AI-powered learning assistant using Python, LangChain, Streamlit, and FastAPI.
- Integrated a locally hosted Llama model with Retrieval-Augmented Generation to answer questions from uploaded PDF documents.
- Implemented document processing, vector embeddings, ChromaDB, and citation-based responses.
- Added features to generate summaries, quizzes, and study materials from course documents.

Smart StudyDesk: AI-Powered Study Organizer

2024

- Designed a cross-platform study application for organizing, summarizing, and reviewing lecture materials.
- Added AI-based note summaries, personalized quizzes, progress tracking, and weak-area analysis.
- Conducted usability testing using semi-structured interviews and NASA-TLX workload analysis.
- Improved the interface based on user feedback, with clearer navigation, folder organization, and feedback visibility.

MarketPulse Website Design and HCI Analysis

2024

- Designed and developed a responsive website for MarketPulse using HTML, CSS, and JavaScript.
- Applied HCI principles including consistency, simplicity, feedback, visibility, constraints, and accessibility.
- Improved the information structure and user interaction through usability-focused design.

Side Hustle Launchpad – Tech4Europe Hackathon, KIT

2024

- Developed a full-stack web application during the Tech4Europe Hackathon at Karlsruhe Institute of Technology, Germany.
- Built analytics dashboards and personalized recommendations to support small-business planning.
- Used FastAPI and MongoDB for backend development and data management.

MOOCS AND RELEVANT COURSEWORK

- **Saarland University:** Data Networks, Neural Networks: Theory and Implementation, Elements of Machine Learning, Human Computer Interaction, High-Level Computer Vision, Generative AI, Machine Translation, Attacks Against Machine Learning Models, and Competitive Programming.
- **University of Bergen:** Functional Programming, Qualitative Analysis, and Philosophy of AI.
- **Certifications:** Machine Learning Specialization – DeepLearning.AI and Stanford Online; CS50's Introduction to Artificial Intelligence with Python – Harvard University; Google Data Analytics Professional Certificate – Google; Python for Data Science, AI & Development – IBM.

TECHNICAL SKILLS AND INTERESTS

Programming : Python, JavaScript, TypeScript, SQL, HTML, CSS, C++, C#, Java

Machine Learning & AI : Scikit-Learn, PyTorch, TensorFlow, Hugging Face Transformers, Large Language Models (LLMs), Prompt Engineering, Retrieval-Augmented Generation (RAG)

LLM Frameworks : LangChain, LangGraph, Ollama, ChromaDB

Web Frameworks : FastAPI, Flask, Streamlit, ReactJS

Libraries & Data : Pandas, NumPy, PySpark, Matplotlib

Tools & Platforms : Git, GitHub Actions, Docker, Microsoft Azure, VS Code, Jupyter, Power BI, Tableau, Figma

Interests : LLM Applications, Agentic AI, AI-Assisted Software Engineering, Data Engineering, Workflow Automation